

Lecture 3

Describing and Summarizing Data

Part 3.

Prices and returns

From Prices to returns

In finance, the raw data are often **prices**: $P_0, P_1, P_2, \dots, P_T$, but the quantity of interest is typically the **return**, because it measures the **relative change** in value from one period to the next.

Returns are more informative than prices when we want to:

- ❑ compare assets with different price levels,
- ❑ measure performance over time,
- ❑ study risk and volatility.

For a price series P_t , the return from period $t - 1$ to t can be defined in two common ways.

Simple return

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

Log return

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

For small returns: $\ln(1 + r_t) \approx r_t$

Compounding return – simple returns

Let r_t be the **simple return** from period $t - 1$ to t , and r_{t+1} the simple return from t to $t + 1$.
The return over the full period from $t - 1$ to $t + 1$ is **not**:

$$\cancel{r_t + r_{t+1}}$$

Instead, simple returns compound as:

$$1 + R_{t-1 \rightarrow t+1} = (1 + r_t)(1 + r_{t+1})$$

$$\text{Therefore: } R_{t-1 \rightarrow t+1} = (1 + r_t)(1 + r_{t+1}) - 1$$

Compounding return – log returns

If r_t and r_{t+1} are **log returns**, then: $r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$ and $r_{t+1} = \ln\left(\frac{P_{t+1}}{P_t}\right)$

The log return over the full period is:

$$R_{t-1 \rightarrow t+1} = \ln\left(\frac{P_{t+1}}{P_{t-1}}\right)$$

Then insert P_t by multiplying and dividing by P_t :

$$R_{t-1 \rightarrow t+1} = \ln\left(\frac{P_{t+1}}{P_{t-1}} \times \frac{P_t}{P_t}\right)$$

Rearrange the terms:

$$R_{t-1 \rightarrow t+1} = \ln\left(\frac{P_t}{P_{t-1}} \times \frac{P_{t+1}}{P_t}\right)$$

Now use the logarithm property:

$$\ln(ab) = \ln(a) + \ln(b)$$

so

$$R_{t-1 \rightarrow t+1} = \ln\left(\frac{P_t}{P_{t-1}}\right) + \ln\left(\frac{P_{t+1}}{P_t}\right)$$

Therefore. $R_{t-1 \rightarrow t+1} = r_t + r_{t+1}$

Annualizing Average Returns

In practice, returns are often observed at a high frequency, for example **daily returns**:

$$r_1, r_2, \dots, r_N$$

We may first compute the **average daily return**:

$$\bar{r}_{\text{daily}} = \frac{1}{N} \sum_{t=1}^N r_t$$

But daily returns are hard to interpret directly, so we often want to express the result as an **annualized average return**.

Annualizing average simple returns

If $\bar{r}_{\text{daily}}$ is an average **simple daily return**, and there are approximately 252 trading days in a year, then:

$$\bar{r}_{\text{annual}} = (1 + \bar{r}_{\text{daily}})^{252} - 1$$

This uses compounding: the average daily return is assumed to be earned repeatedly over the year.

More generally, if there are m periods per year:

$$\bar{r}_{\text{annual}} = (1 + \bar{r}_{\text{period}})^m - 1$$

Annualizing Average log returns

If $\bar{r}_{\text{daily}}$ is an average **log daily return**, then annualization is simpler:

$$\bar{r}_{\text{annual}} = 252 \times \bar{r}_{\text{daily}}$$

More generally, if there are m periods per year :

$$\bar{r}_{\text{annual}} = m \times \bar{r}_{\text{period}}$$

because log returns are additive over time.

Annualizing Variance

Returns are often computed at a high frequency, for example daily returns:

$$r_1, r_2, \dots, r_N$$

After computing the **daily variance** and **daily standard deviation**, we often want to express risk on an **annual scale**.

Annualizing variance

If σ_{daily}^2 is the variance of daily returns, then:

$$\sigma_{\text{annual}}^2 = 252 \times \sigma_{\text{daily}}^2$$

More generally, if there are m periods per year

$$\sigma_{\text{annual}}^2 = m \times \sigma_{\text{period}}^2$$

We are assuming that returns are approximately independent over time and measured at the same frequency.

Annualizing Volatility

Annualizing standard deviation / volatility

Standard deviation is the square root of variance, so:

$$\sigma_{\text{annual}} = \sqrt{\sigma_{\text{annual}}^2}$$

Therefore:

$$\sigma_{\text{annual}} = \sqrt{252} \times \sigma_{\text{daily}}$$

More generally:

$$\sigma_{\text{annual}} = \sqrt{m} \times \sigma_{\text{period}}$$

Quick Exercise: Annualizing Return and Volatility

Suppose a stock has the following **average daily return** and **daily volatility**:

$$\bar{r}_{daily} = 0.05\% = 0.0005$$

$$\sigma_{daily} = 1.2\% = 0.012$$

Assume there are: $m = 252$ trading days in one year.

Questions

1. Compute the **annualized average simple return**:

2. Compute the **annualized volatility**:

3. Compare the exact annualized return with the linear approximation:

$$\bar{r}_{annual} \approx 252 \times \bar{r}_{daily}$$

Solution. Annualizing Return and Volatility

1. Exact annualized average return

$$\bar{r}_{\text{annual}} = (1 + 0.0005)^{252} - 1$$

$$\bar{r}_{\text{annual}} \approx 0.1342$$

$$\bar{r}_{\text{annual}} \approx 13.42\%$$

2. Annualized volatility

$$\sigma_{\text{annual}} = \sqrt{252} \times 0.012$$

$$\sigma_{\text{annual}} \approx 0.1905$$

$$\sigma_{\text{annual}} \approx 19.05\%$$

3. Linear approximation

$$\bar{r}_{\text{annual}} \approx 252 \times 0.0005$$

$$\bar{r}_{\text{annual}} \approx 0.126$$

$$\bar{r}_{\text{annual}} \approx 12.60\%$$

Since the exact compounded return is: 13.42% the difference is **0.82%**

The approximation is reasonably close, but it slightly underestimates the annualized return because it ignores compounding. It works well for small daily returns, but becomes less accurate as the return per period increases or as the number of compounding periods becomes larger.